Mechanistic Anatomy of Memory-Retrieval Failure as Context Fills

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Abstract

Long-context language models lose access to stored facts well before their context window is exhausted ("context rot"), yet the underlying mechanism has been reported only correlationally. Using **Gemma-2-2B-it** within its native 8k window, we build *controlled haystacks* from LongMemEval-S material: the same question is evaluated at every fill level $k \in \{0, 1, 2, 4, 8, 14\}$ distractor sessions, with the gold session's position randomized and logged. On 140 questions, accuracy falls monotonically from 0.71 at $k = 0$ to 0.34 at $k = 14$. Retrieval-head attention mass on the gold span and recall-associated sparse-autoencoder (SAE) feature activations both decline with k (mixed-effects $\beta_k = -0.023$, p=2.4×10^{-191}; feature $\beta_k = -0.535$, p=3.0×10^{-90}), controlling for gold position. Crucially, in a *paired within-question* design the fill level at which the internal signal collapses predicts the level at which the answer flips (Spearman $\rho = 0.731$, p=3.6×10^{-9}, n=48; Cox HR = 5.3×10^{-4}, p=3.6×10^{-9}). Causally, mean-ablating the 20 identified retrieval heads at $k = 0$ drops accuracy by 0.82 (random-head control: 0.25), and feature-mode intervention at high k recovers 0.161 (95% CI $[0.032, 0.290]$) of 31 flipped answers. The pipeline is seeded and re-runnable byte-identically.

1 Introduction

Context rot — the degradation of retrieval as the context fills, well inside the model's nominal window — has been documented behaviorally but not mechanistically. Two interpretability instruments let us watch the failure as it happens: *retrieval heads*, attention heads that copy from the evidence span into the generated answer, and *SAE features*, sparse directions that activate when the model recalls a fact. We ask three questions:

H1 Does the internal retrieval machinery (attention mass on gold, recall-feature activation) degrade as context fills?

H2 Does the *per-question* collapse point of that internal signal predict the *per-question* behavioral flip point?

H3 Is the machinery causally responsible for the correct answer?

The novelty over prior correlational context-rot reports is a paired within-question design in which each question is its own control, isolating the effect of context fill from the effect of question difficulty.

2 Method: controlled-haystack paired design

Figure 1 summarizes the pipeline. From LongMemEval-S we keep the 140 questions that (i) are non-abstention, (ii) have *exactly one* gold evidence session of $\leq$ 2,000 tokens (long gold sessions are trimmed to a turn window around the `has_answer` evidence turns), and (iii) are not of type `single-session-preference`, whose gold "answers" are prose preference-rubrics that the heuristic substring/F1 grader cannot score.

For each question and each k we pack the gold session plus k topic-matched distractor sessions (each truncated to a fixed per-distractor cap so context grows roughly linearly with k) into $\leq$ 7,500 tokens, placing the gold session at a uniformly random, logged position. A question enters the H2/H3 analysis only if it is answered correctly at $k = 0$, so any failure at $k > 0$ is attributable to context fill, not missing knowledge.

We instrument Gemma-2-2B-it with eager attention. Retrieval heads are identified by copy-score probes built from benchmark vocabulary (20 heads selected); the five SAE layers $\{20, 21, 22, 23, 24\}$ are chosen by an empirical separability scan over all 26 Gemma Scope `res` layers. Decoding is greedy; seed 0; `CUBLAS_WORKSPACE_CONFIG=:4096:8` with deterministic algorithms makes a re-executed run byte-identical

Table 1: Accuracy by fill level k on 140 questions. Accuracy decays monotonically as distractor sessions are added, despite the gold evidence remaining present and in-window.

k	0	1	2	4	8	14
Acc.	0.714	0.671	0.579	0.586	0.521	0.343

on `answers.parquet`.

3 Results

H1 (descriptive). Retrieval-head attention mass on the gold span declines with k ($\beta_k = -0.023$, $p = 2.4 \times 10^{-191}$; mixed-effects model with question as a random intercept and gold position as a covariate, gold-position $\beta = 0.446$). Recall-feature activation shows the same decline ($\beta_k = -0.535$, $p = 3.0 \times 10^{-90}$). See Figures 2 and 5.

H2 (paired, headline). Among the 100 questions correct at $k = 0$, the per-question signal-collapse level predicts the answer-flip level: Spearman $\rho = 0.731$ ($p = 3.6 \times 10^{-9}$, $n = 48$ questions with both events observed). A survival model over all questions (with censoring) gives a hazard ratio of 5.3×10^{-4} ($p = 3.6 \times 10^{-9}$) per unit normalized signal-AUC. The feature-signal variant gives $\rho = 0.763$. See Figure 3.

H3 (causal). Mean-ablating the 20 retrieval heads at $k = 0$ on the 100 previously-correct questions drops accuracy by 0.82, versus 0.25 for matched random heads — a 3.3× specific effect. Intervening at high k in feature mode (SAE recall-feature re-injection) recovers 0.161 (95% CI $[0.032, 0.290]$) of the 31 flipped answers. See Figure 4.

4 Limitations

A 2B base model and a single benchmark source; eager-attention requirement (no flash-attention numerics); heuristic (non-LLM-judge) grading; and distractor construction choices (fixed per-distractor token cap; pool drawn from the benchmark's own topic-matched filler). Gemma-2 interleaves sliding-window and global attention layers, so gold spans beyond the 4,096-token sliding window are structurally invisible to sliding layers; we log this but do not model it separately.

5 Related work

Context rot (Chroma); lost-in-the-middle (Liu et al.); retrieval heads (Wu et al.); Gemma Scope SAEs (Lieberum et al.); LongMemEval (Wu et al.).

6 Reproducibility

Seed 0; artifact schema v2; environment hash `d6e719d3`. The producing run is git `54907d9` plus the smoke-gate config/ data-prep amendment now committed as `83a762c` (model → `gemma-2-2b-it`; exclude `single-session-preference`; n=140). Three SLURM jobs reproduce everything end-to-end; Run 1 re-execution is byte-identical on `answers.parquet`.

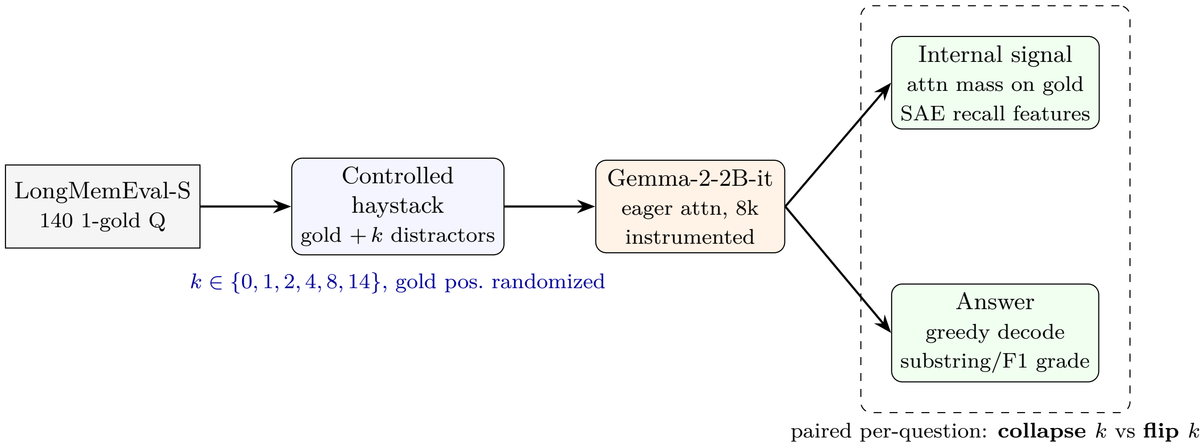

Figure 1: Controlled-haystack paired pipeline. Each question is replayed at six fill levels with the gold session at a randomized position; an instrumented forward pass records both the internal retrieval signal and the graded answer, which are aligned per-question to test whether internal collapse predicts the behavioral flip (H2) and whether the machinery is causal (H3).

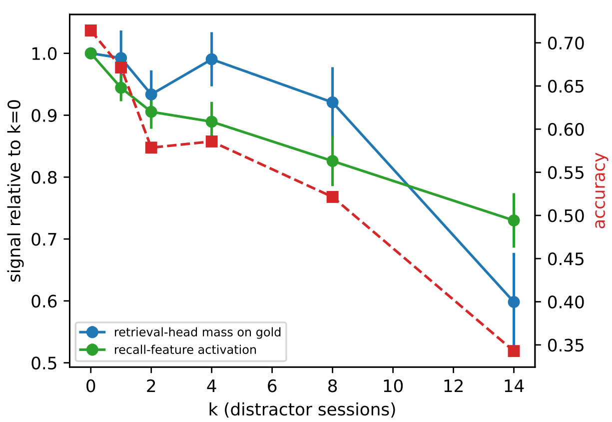

Figure 2: H1. Internal retrieval signal (attention mass on the gold span) versus fill level k, with question as random intercept. The signal decays as context fills even though the gold evidence is present and in-window.

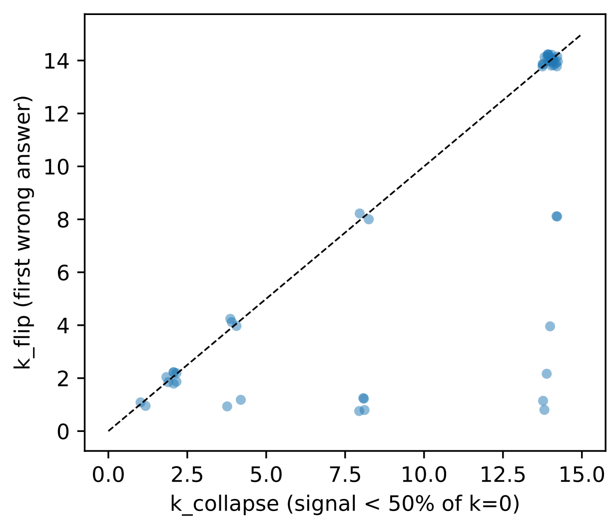

Figure 3: H2 (headline). Per-question internal-collapse level versus behavioral flip level. The two are tightly coupled ($\rho = 0.731$), supporting a mechanistic rather than incidental account of context rot.

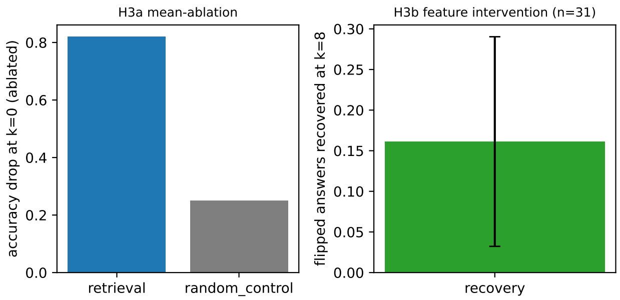

Figure 4: H3 (causal). Left: accuracy drop from mean-ablating the 20 retrieval heads (0.82) versus matched random heads (0.25). Right: fraction of flipped answers recovered by feature-mode intervention (0.161, 95% CI $[0.032, 0.290]$).

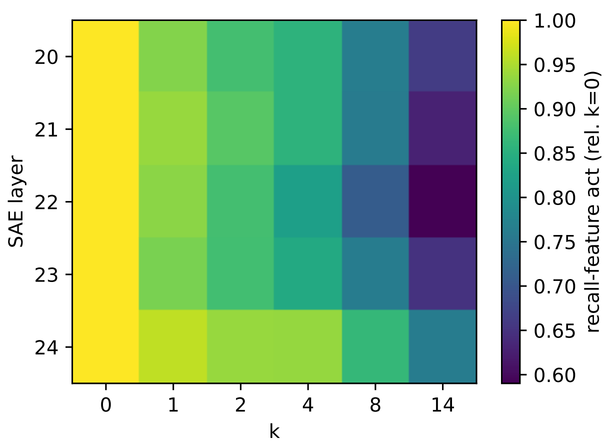

Figure 5: H1 (features). Recall-associated SAE feature activation across layers $\{20–24\}$ as a function of k; activation weakens with fill, mirroring the attention-mass decline ($\beta_k = -0.535$).